The limiting shift in Nemes's Gamma-function expansion

MathDB 319738: full solution

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Task 1. Let g_n, v_n be determined by matching successive powers of x^{-1} in

$$\frac{\Gamma(x+1)e^x}{x^x\sqrt{2\pi(x+1/6)}} \sim 1 + \sum_{n\geq 1} \frac{g_n}{(x+v_n)^{2n}}. \tag{1}$$

Prove that $v_n \to 1/4$. This is Conjecture 2.2 in Nemes's published paper [1], p. 24; its defining recurrences are (2.11)–(2.12) on that page. The first shifts are 23/90 and 1792627/7289730.

Theorem 1. *The coefficient matching in* (1) *determines real g_n, v_n for every $n \geq 1$, with $g_n \neq 0$, and*

$$\lim_{n\to\infty} v_n = \frac{1}{4}.$$

In fact, for every $n \geq 1$,

$$\left| v_n - \frac{1}{4} \right| < \frac{1}{40\pi}, \qquad \frac{1}{4} \leq \frac{(2\pi)^{2n}|g_n|}{(2n-1)!} \leq \frac{9}{25}.$$

Moreover,

$$g_n \sim \frac{(-1)^{n+1}(2n-1)!}{\pi(2\pi)^{2n}}.$$

All series manipulations below are formal manipulations of asymptotic coefficients. Convergence of the asymptotic series is not required. The norm introduced below controls their factorial growth.

Centering at 1/4. Put $y = x + 1/4$, $\rho = 2\pi$, and let

$$F(y) = \frac{\Gamma(y+3/4)e^{y-1/4}}{(y-1/4)^{y-1/4}\sqrt{2\pi(y-1/12)}} \sim 1 + \sum_{m\geq 2} a_m y^{-m}.$$

Write $\log F(y) \sim \sum_{m\geq 2} \lambda_m y^{-m}$. The shifted Stirling expansion [2, Eq. 5.11.8] gives

$$\lambda_m = \frac{(-1)^{m+1}B_{m+1}(3/4) - 4^{-(m+1)}}{m(m+1)} + \frac{12^{-m}}{2m} \qquad (m \geq 1). \tag{2}$$

Here $B_j(u)$ is the Bernoulli polynomial. To check the elementary terms, subtract the shifted Stirling expansion from $\log F(y)$; the difference is

$$-(y-1/4)\log(1-1/(4y)) - \tfrac{1}{2}\log(1-1/(12y)) - \tfrac{1}{4}.$$

In particular, $\lambda_1 = 0, \lambda_2 = 1/144, \lambda_3 = -1/12960$. Define normalized coefficients

$$\ell_m = \frac{\rho^m \lambda_m}{(m-1)!}, \qquad A_m = \frac{\rho^m a_m}{(m-1)!} \quad (m \geq 2).$$

Lemma 1. *For $n \geq 1$,*

$$\frac{27}{100} < |\ell_{2n}| < \frac{1}{3}, \qquad |\ell_{2n+1}| < \frac{1}{100}.$$

Also $\ell_{2n+1} \to 0$ and $\ell_{2n} - (-1)^{n+1}/\pi \to 0$.

Proof. The Bernoulli Fourier series [2, Eqs. 24.8.1–24.8.2] give, with $\beta(s) = \sum_{j\geq 0}(-1)^j(2j+1)^{-s}$,

$$\ell_{2n} = \frac{(-1)^{n+1}}{\pi}\beta(2n+1) + e_{2n}, \tag{3}$$

$$\ell_{2n-1} = \frac{(-1)^n}{\pi\, 2^{2n}}(1 - 2^{1-2n})\zeta(2n) + e_{2n-1}, \tag{4}$$

$$e_m = -\frac{(\pi/2)^m}{4(m+1)!} + \frac{(\pi/6)^m}{2m!}. \tag{5}$$

For $m \geq 2$, $e_m \leq 0$ and

$$|e_m| \leq R_m := \frac{(\pi/2)^m}{4(m+1)!}.$$

Indeed, the ratio of the positive term to the negative term in (5) is $2(m+1)/3^m \leq 1$. The sequence R_m is decreasing.

For $m = 2$, $\ell_2 = \pi^2/36$, which has the asserted bounds. For even $m \geq 4$, use $1 - 3^{-(m+1)} < \beta(m+1) < 1$ and $157/50 < \pi < 22/7$. The following rational bounds suffice:

$$\frac{7}{22}(1 - 3^{-5}) - \frac{(11/7)^4}{480} > \frac{27}{100}, \qquad \frac{50}{157} + \frac{(11/7)^4}{480} < \frac{1}{3}.$$

For $m = 3$, $\ell_3 = -\pi^3/3240$, of absolute value less than $1/100$. For odd $m \geq 5$, the estimate $\zeta(m+1) \leq 1 + 1/m$, obtained by integral comparison, gives

$$|\ell_m| \leq \frac{1 + 1/m}{\pi\, 2^{m+1}} + R_m \leq \frac{50}{157}\frac{6}{5 \cdot 64} + \frac{(11/7)^5}{2880} < \frac{1}{100}.$$

The same formulas prove both limits. □

Passing from the logarithm to the function. For a formal series $P(z) = \sum_{m\geq 2} p_m z^m$, set

$$\|P\| = \sup_{m\geq 2} \frac{\rho^m |p_m|}{(m-1)!}.$$

If P, Q have finite norm, then

$$\frac{\rho^m |[z^m]PQ|}{(m-1)!} \leq W_m \|P\| \|Q\|, \qquad W_m = \sum_{j=2}^{m-2} \frac{(j-1)!(m-j-1)!}{(m-1)!}. \tag{6}$$

Empty sums are zero. Directly, $W_4 = W_5 = 1/6$ and $W_6 = 2/15$. For $m \geq 7$,

$$W_m = \frac{1}{m-1}\sum_{j=2}^{m-2} \binom{m-2}{j-1}^{-1} \leq \frac{1}{m-1} \leq \frac{1}{6}.$$

Consequently, $W_m \to 0$ and

$$\|PQ\| \leq \tfrac{1}{6} \|P\| \|Q\|. \tag{7}$$

Lemma 2. *For every $n \geq 1$,*

$$\frac{13}{50} < |A_{2n}| < \frac{7}{20}, \qquad |A_{2n+1}| < \frac{11}{1000}.$$

Furthermore, $A_{2n+1} \to 0$ and $A_{2n} - (-1)^{n+1}/\pi \to 0$.

Proof. Let $L(z) = \sum_{m\geq 2} \lambda_m z^m$. Since $\|L\| \leq 1/3$, (7) gives

$$\left\|e^L - 1 - L\right\| \leq 6(e^{1/18} - 1 - 1/18) \leq \frac{1}{106} < \frac{1}{100}.$$

The penultimate bound follows by bounding the tail of the exponential series by a geometric series with ratio $1/54$. Together with Lemma 1, this proves the even bounds.

For the sharper odd bound, split $L = E + O$ into its even and odd parts. Then $\|E\| \leq 1/3$, $\|O\| \leq 1/100$. The odd coefficients of e^L are those of $e^E(e^O - 1)$. By (7),

$$\left\|e^E(e^O - 1)\right\| \leq 6e^{1/18}(e^{1/600} - 1) \leq \frac{216}{20383} < \frac{11}{1000}.$$

For the rational estimate, use $e^{1/18} \leq 18/17$ and $e^{1/600} - 1 \leq 2/1199$, again by geometric majorants.

Finally, applying (6) once and (7) to the remaining factors in each L^k shows that

$$|A_m - \ell_m| \leq W_m \sum_{k\geq 2} \frac{(1/3)^k (1/6)^{k-2}}{k!} \leq \frac{W_m}{18} e^{1/18} \longrightarrow 0.$$

Lemma 1 now gives the limits. □

The normalized triangular recurrence. Set

$$c_n = \frac{\rho^{2n} g_n}{(2n-1)!}, \qquad t_n = \rho(v_n - 1/4), \qquad q = \frac{1}{20}.$$

Expanding each term of (1) in powers of y^{-1} and dividing the coefficients by their factorials gives exactly

$$c_n = A_{2n} - \sum_{j=1}^{n-1} c_j \frac{t_j^{2n-2j}}{(2n-2j)!}, \tag{8}$$

$$c_n t_n = -A_{2n+1} - \sum_{j=1}^{n-1} c_j \frac{t_j^{2n+1-2j}}{(2n+1-2j)!}. \tag{9}$$

For clarity, the elementary cancellation behind these formulas is

$$\frac{\rho^{2j+r}}{(2j+r-1)!} \binom{2j+r-1}{r} g_j (v_j - 1/4)^r = c_j \frac{t_j^r}{r!}.$$

The sign in (9) comes from the odd exponent in $(1 + (v_j - 1/4)/y)^{-2j}$.

We prove simultaneously by induction that the recurrence is defined and that

$$\frac{1}{4} \leq |c_n| \leq \frac{9}{25}, \qquad |t_n| < q \quad (n \geq 1). \tag{10}$$

Assume these assertions for $j < n$; the same estimates apply when $n = 1$, with empty sums. From (8),

$$|c_n - A_{2n}| \le \frac{9}{25}(\cosh q - 1) < \frac{9}{25 \cdot 790} < \frac{1}{100}.$$

Lemma 2 therefore puts $|c_n|$ between $1/4$ and $9/25$; in particular, division by c_n is legitimate. Now (9) gives

$$|t_n| \le 4\left(\frac{11}{1000} + \frac{9}{25}(\sinh q - q)\right) < 4\left(\frac{11}{1000} + \frac{9}{25 \cdot 47000}\right) < q.$$

The elementary estimates used here are

$$\cosh(1/20) - 1 \le \frac{6}{4799} < \frac{1}{790}, \qquad \sinh(1/20) - 1/20 \le \frac{1}{47994} < \frac{1}{47000}.$$

They follow by bounding the respective positive series tails by geometric series. This completes the induction and proves existence of every shift, as well as the global bounds.

Convergence. Using (10) in (9), with $r = n - j$, yields

$$|t_n| \le 4|A_{2n+1}| + \frac{36}{25}\sum_{r=1}^{n-1} \frac{q^{2r}}{(2r+1)!}|t_{n-r}|. \tag{11}$$

The weights on the right have total mass

$$\kappa = \frac{36}{25}\left(\frac{\sinh q}{q} - 1\right) \le \frac{36}{25}(\cosh q - 1) < 1.$$

Let $M = \limsup_n |t_n|$, which is finite by (10). Lemma 2 and (11) imply $M \le \kappa M$. One can justify this last passage by first retaining finitely many weights, taking the limsup, and then bounding the omitted tail using $|t_j| \le q$. Hence $M = 0$, and $v_n \to 1/4$.

For the final assertion of Theorem 1, (8) and dominated convergence for the summable majorant $(9/25)q^{2r}/(2r)!$ give $c_n - A_{2n} \to 0$. The even limit in Lemma 2 proves $c_n - (-1)^{n+1}/\pi \to 0$, completing the proof. □

Task 2. Determine the first nonzero asymptotic correction to $v_n = 1/4$, including its sign and an explicit remainder bound.

Task 3. Extend the argument to suitably normalized shifted Gamma functions $\Gamma(x + a)$, and determine which limiting shifts occur.

Task 4. Derive a useful bound for $|v_n - 1/4|$ in terms of n, with explicit constants valid for every $n \ge 1$.

Task 5. Combine the coefficient bounds with asymptotic remainder estimates for (1) to obtain a certified evaluation algorithm.

References

[1] G. Nemes, *More accurate approximations for the Gamma function*, Thai J. Math. **9** (2011), 21–28. https://thaijmath.in.cmu.ac.th/index.php/thaijmath/article/view/222. Preprint: https://arxiv.org/abs/1003.6020.

[2] NIST Digital Library of Mathematical Functions, https://dlmf.nist.gov/5.11.E8, https://dlmf.nist.gov/24.8.E1, and https://dlmf.nist.gov/24.8.E2.